Geodesic Optimality in Pandrosion Probe Geometry

Finite-slope probes as low-action paths in a stability metric

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Abstract

This paper proposes that optimal Pandrosion probes are geodesic objects. We define a stability metric built from residual, conditioning, logarithmic energy, and equal-value pole barriers. We prove a local Euclidean limit and a finite-action pole avoidance theorem, then state a geodesic optimality conjecture for probe engines.

Scope and status. This manuscript is written as a theorem-and-conjecture paper inside the Pandrosion research programme. It gives precise definitions, conditional theorems, and conjectures that can be tested by the existing finite-slope engines. The results are structural: they are not trading models and do not rely on empirical market data.

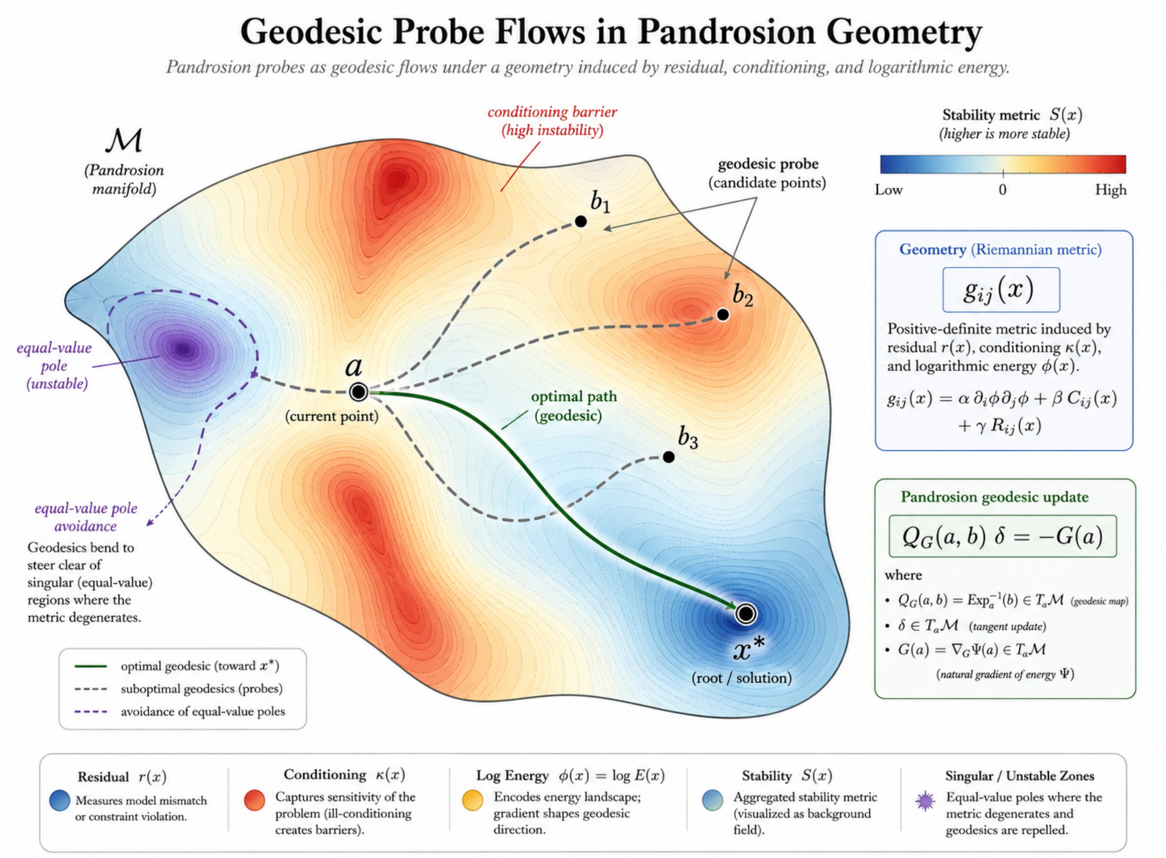

Figure 1: Conceptual overview for this paper. The diagram is intentionally synthetic: it organizes the geometric objects used in the definitions and conjectures.

1 Probe manifolds and metric costs

A Pandrosion update depends on a probe b. Instead of regarding b as a heuristic sample, we regard it as an endpoint of a curve γ starting at the current point a. The cost of the curve is determined by a stability metric.

Definition 1.1 (Pandrosion stability metric). *Let $r(x) = \|G(x)\|$, let $\kappa(x)$ denote a local finite-slope conditioning proxy, and let $\phi(x) = \log(1 + \|x\|)$. A stability metric is a positive definite tensor of the form*

$$g_x = \alpha\, \mathrm{d}\phi \otimes \mathrm{d}\phi + \beta\, C_x + \gamma\, R_x + \lambda\, P_x, \tag{1}$$

where C_x penalizes conditioning, R_x penalizes residual curvature, and P_x is an equal-value pole barrier.

The action of a probe path is

$$\mathcal{S}(\gamma) = \int_0^1 \sqrt{g_{\gamma(t)}(\dot\gamma(t), \dot\gamma(t))}\, \mathrm{d}t. \tag{2}$$

Optimal probes are endpoints of low-action curves.

2 Local geodesic limit

When the metric is flat and the singular barriers are absent, the geodesic viewpoint reduces to the usual finite-slope probe.

Theorem 2.1 (Euclidean local limit). *Assume $g_x = g_0 + O(\|x - a\|)$ near a and no pole barrier is active. Then the minimizing probe path from a to b is, to first order, the straight segment*

$$\gamma(t) = a + t(b - a) + O(\|b - a\|^2). \tag{3}$$

Consequently the geodesic finite-slope matrix agrees with the ordinary telescopic finite-slope matrix up to second-order path error.

Proof. In normal coordinates for g_0, geodesics satisfy $\ddot\gamma^k + \Gamma^k_{ij}\dot\gamma^i\dot\gamma^j = 0$. Since $\Gamma = O(\|x - a\|)$, the first-order solution is the straight line. The finite-slope integral along the geodesic differs from the coordinate segment integral only by the quadratic path displacement. □

3 Pole barriers

Equal-value pole sets have the form

$$\mathcal{P} = \{(a, b) : G(a) = G(b)\}. \tag{4}$$

When projected into the probe manifold, they form regions where finite-slope information degenerates. The metric should become expensive near these regions.

Theorem 3.1 (Finite-action pole avoidance). *Assume that near an equal-value pole hypersurface Π the metric contains a barrier term*

$$P_x \succeq \frac{c}{\mathrm{dist}(x, \Pi)^2} I. \tag{5}$$

Then every path crossing Π has infinite action. Therefore every finite-action geodesic avoids Π.

Proof. Let $s(t) = \mathrm{dist}(\gamma(t), \Pi)$. If γ crosses Π, then $s(t_0) = 0$ for some t_0. The barrier contribution to the path length dominates $\int |\dot s|/s\, dt$ near t_0, which diverges logarithmically. Hence finite action is impossible for crossing paths. □

4 Geodesic probe conjectures

Conjecture 4.1 (Geodesic optimality conjecture)**.** *Let b_k be the probe selected by a probe engine minimizing a composite score made of residual decrease, finite-slope conditioning, logarithmic energy, and equal-value separation. In the small-step limit, the piecewise linear paths from a_k to b_k converge to geodesics of a stability metric g.*

Conjecture 4.2 (Geodesic descent conjecture)**.** *If a Pandrosion direction is built from a finite-action geodesic probe and the associated finite-slope matrix is uniformly coercive, then the resulting update is energy-decreasing for sufficiently small flow time.*

5 Algorithmic implications

The geodesic perspective suggests that probes should not merely be sampled radially. A robust engine should search for low-cost paths in the stability metric. Practical approximations include: (i) detouring around equal-value pole barriers; (ii) penalizing directions that cross high-conditioning ridges; (iii) preferring probes whose logarithmic energy evolves smoothly; and (iv) applying spectral filtering when the local metric becomes anisotropic.

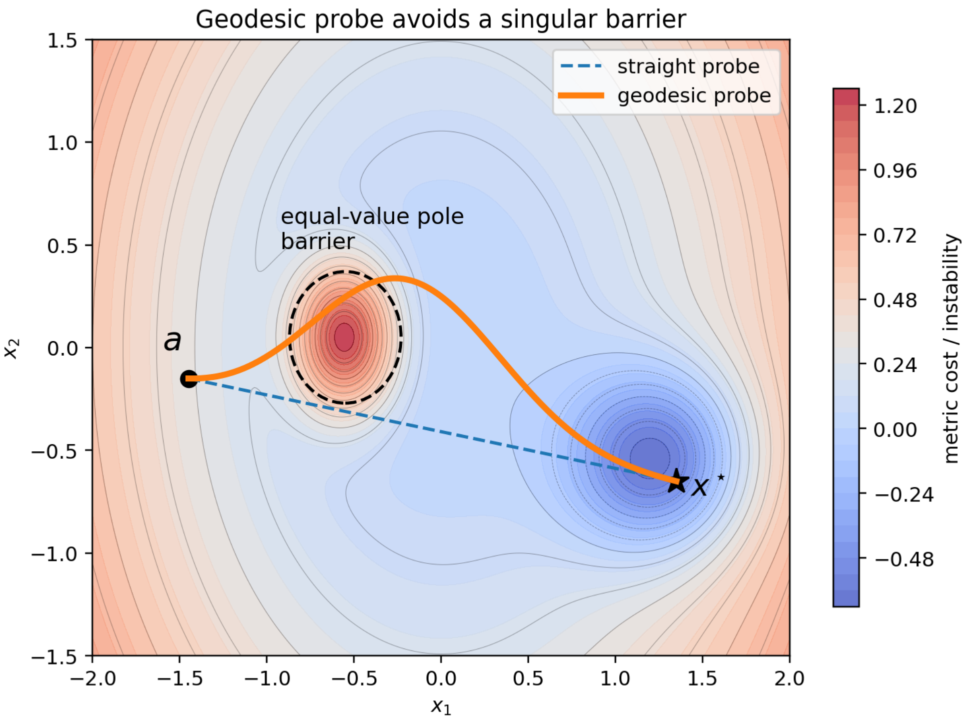

Figure 2: A supporting numerical-geometric schematic generated from the formal objects of the paper.

6 Outlook

The main purpose of this note is to isolate a precise mathematical direction. The next step is to attach each conjecture to a benchmarkable engine: finite-slope descent tests for the first paper, chart-selection tests for the second, and probe-path tests for the third. A successful theory should explain not only convergence, but also why the equal-value poles, logarithmic normalization, and projective charts observed in prior Pandrosion experiments are structurally necessary.

References

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